1. hard process (ME): $q\bar{q} \rightarrow b\bar{b}$

run 1 lumi 1 event 1

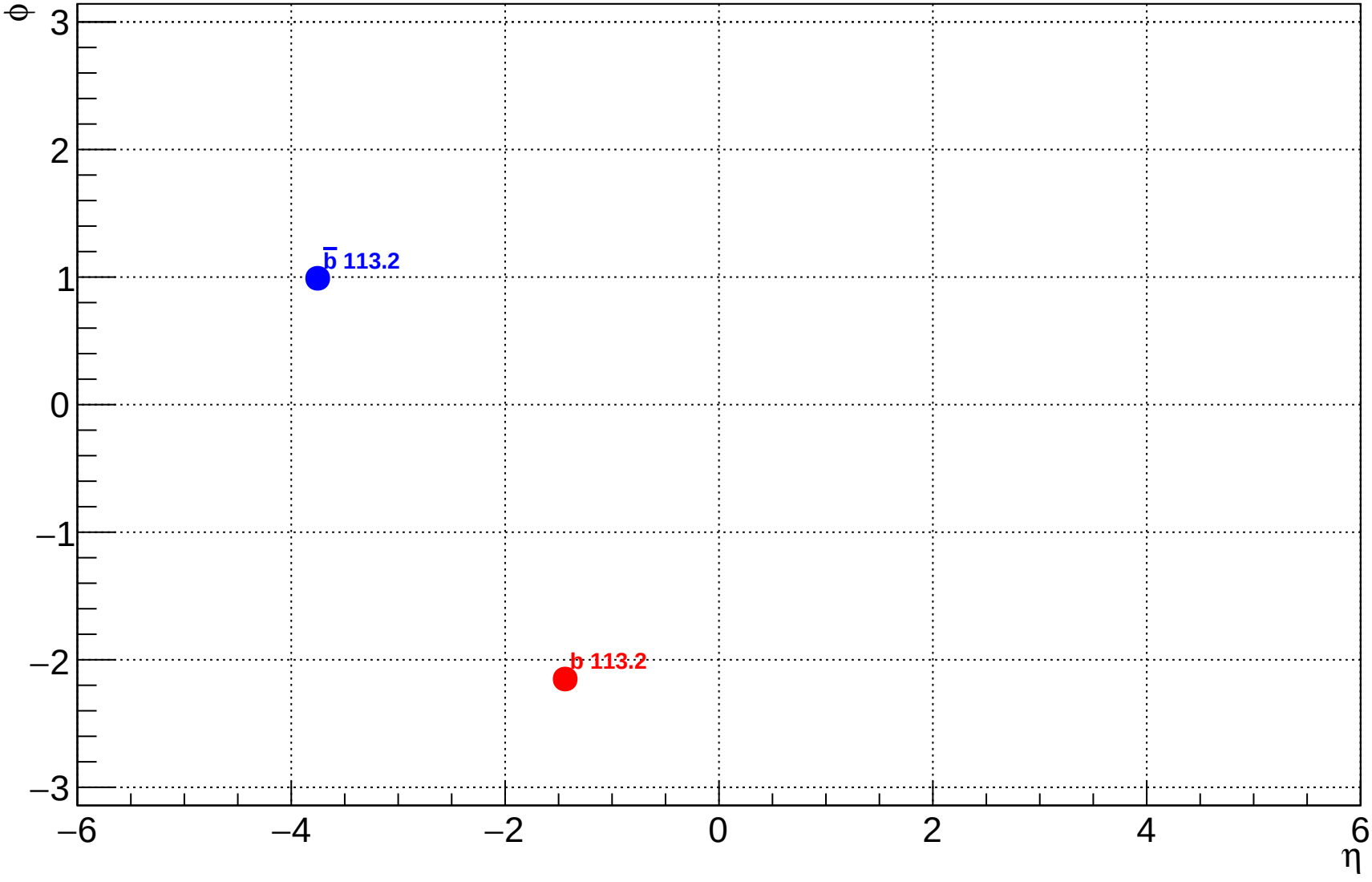
2 particles drawn (marker size ~ log p_T; label = p_T [GeV] for p_T > 5 GeV)

b quark (status 23): p_T = 113.22 GeV, η = -1.44, ϕ = -2.15

$\bar{b}$ quark (status 23): p_T = 113.22 GeV, η = -3.75, ϕ = 0.99

net b+$\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 0.00 GeV (p_x = -0.00, p_y = 0.00), $\Sigma|p_T|$ = 226.44 GeV, $\Delta\phi(b,\bar{b})$ = 3.14

sample ME_PS_MPI_Had
ISR on, FSR on, MPI on
Had on, Decay off
pTHatMin 100 GeV



particle origin (number drawn)

- b system (1)
- $\bar{b}$ system (1)
- b + $\bar{b}$ (vector sum)

marker size:

- p_T = 1 GeV
- p_T = 10 GeV
- p_T = 100 GeV

stages:

1 hard process
2 ISR + FSR
3 MPI / remnants
4 hadronisation
5 hadron decays

2. + parton shower (ISR + FSR)

run 1 lumi 1 event 1

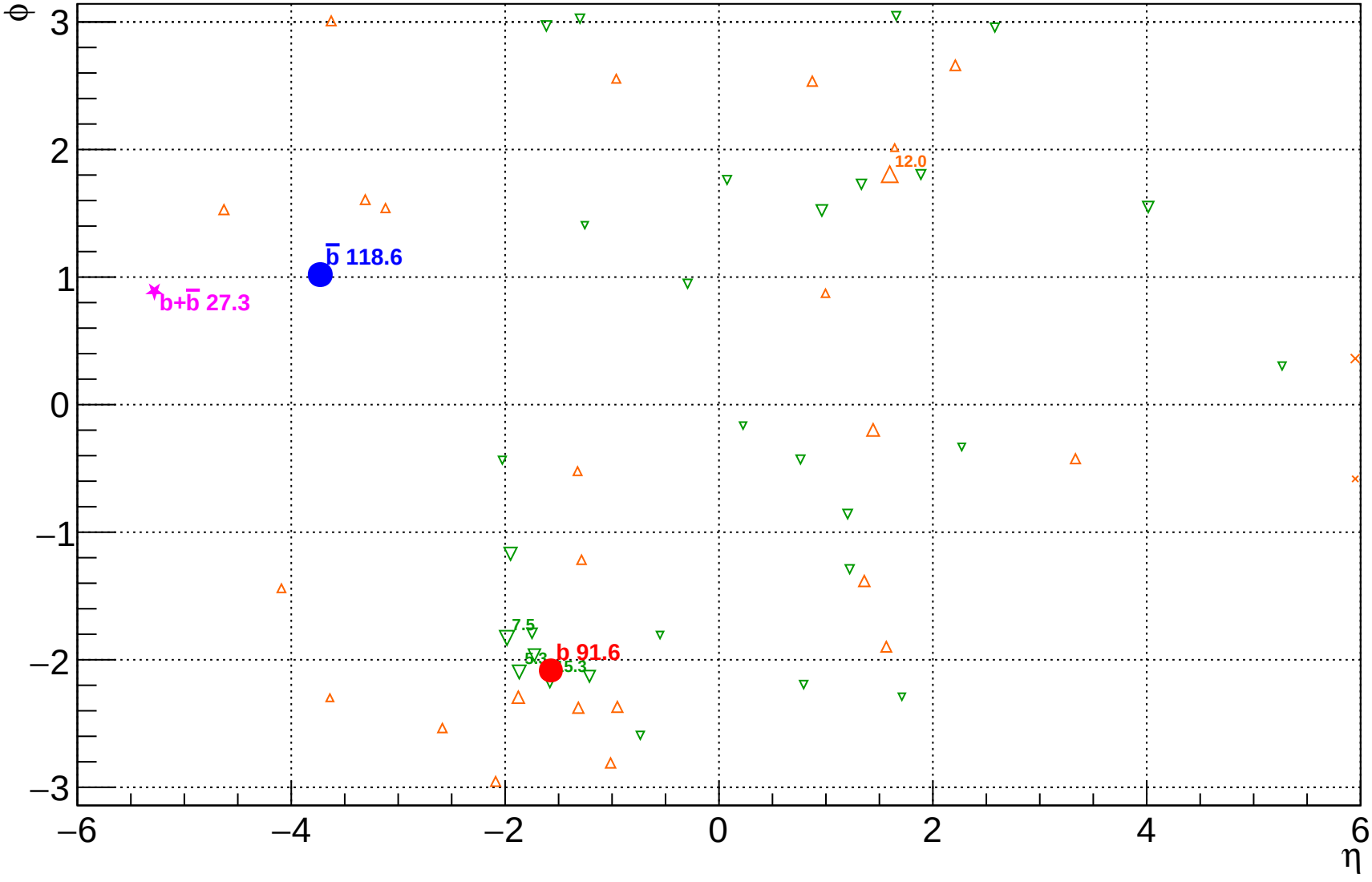
57 particles drawn (marker size ~ log p_T; label = p_T [GeV] for p_T > 5 GeV), 2 with |η| > 6 drawn on the border (x)

b quark (status 52): p_T = 91.59 GeV, η = -1.57, φ = -2.08

$\bar{b}$ quark (status 52): p_T = 118.57 GeV, η = -3.73, φ = 1.02

net b+$\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 27.29 GeV (p_x = 17.19, p_y = 21.19), $\Sigma|p_T|$ = 210.16 GeV, $\Delta\phi(b,\bar{b})$ = 3.10

sample ME_PS_MPI_Had
ISR on, FSR on, MPI on
Had on, Decay off
pTHatMin 100 GeV



particle origin (number drawn)

- b system (1)
- $\bar{b}$ system (1)
- b + $\bar{b}$ (vector sum)
- ISR (26)
- FSR (29)

marker size:

- p_T = 1 GeV
- p_T = 10 GeV
- p_T = 100 GeV

stages:

1 hard process
2 ISR + FSR
3 MPI / remnants
4 hadronisation
5 hadron decays

3. + MPI / beam remnants

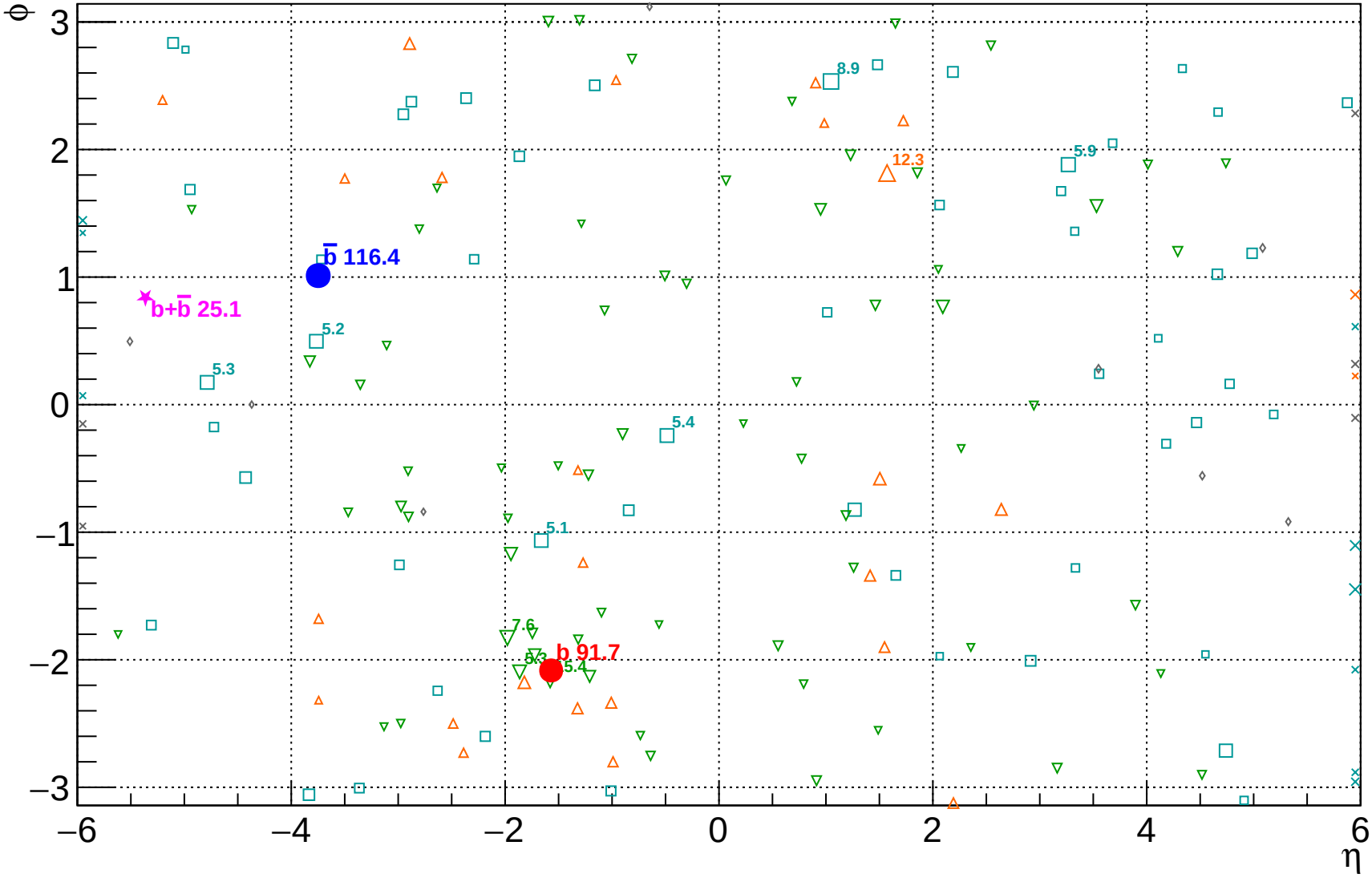
run 1 lumi 1 event 1

169 particles drawn (marker size ~ log p_T; label = p_T [GeV] for p_T > 5 GeV), 16 with |η| > 6 drawn on the border (x)

b quark (status 62): p_T = 91.74 GeV, η = -1.57, φ = -2.08

$\bar{b}$ quark (status 62): p_T = 116.36 GeV, η = -3.75, φ = 1.01

net b+$\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 25.08 GeV (p_x = 16.73, p_y = 18.68), $\Sigma|p_T|$ = 208.10 GeV, $\Delta\phi(b,\bar{b})$ = 3.10



sample ME_PS_MPI_Had
ISR on, FSR on, MPI on
Had on, Decay off
pTHatMin 100 GeV

particle origin (number drawn)

- b system (1)
- $\bar{b}$ system (1)
- b + $\bar{b}$ (vector sum)
- ISR (26)
- FSR (67)
- MPI (61)
- beam remnant (13)

marker size:

- p_T = 1 GeV
- p_T = 10 GeV
- p_T = 100 GeV

stages:

1 hard process
2 ISR + FSR
3 MPI / remnants
4 hadronisation
5 hadron decays

4. + hadronisation (primary hadrons)

run 1 lumi 1 event 1

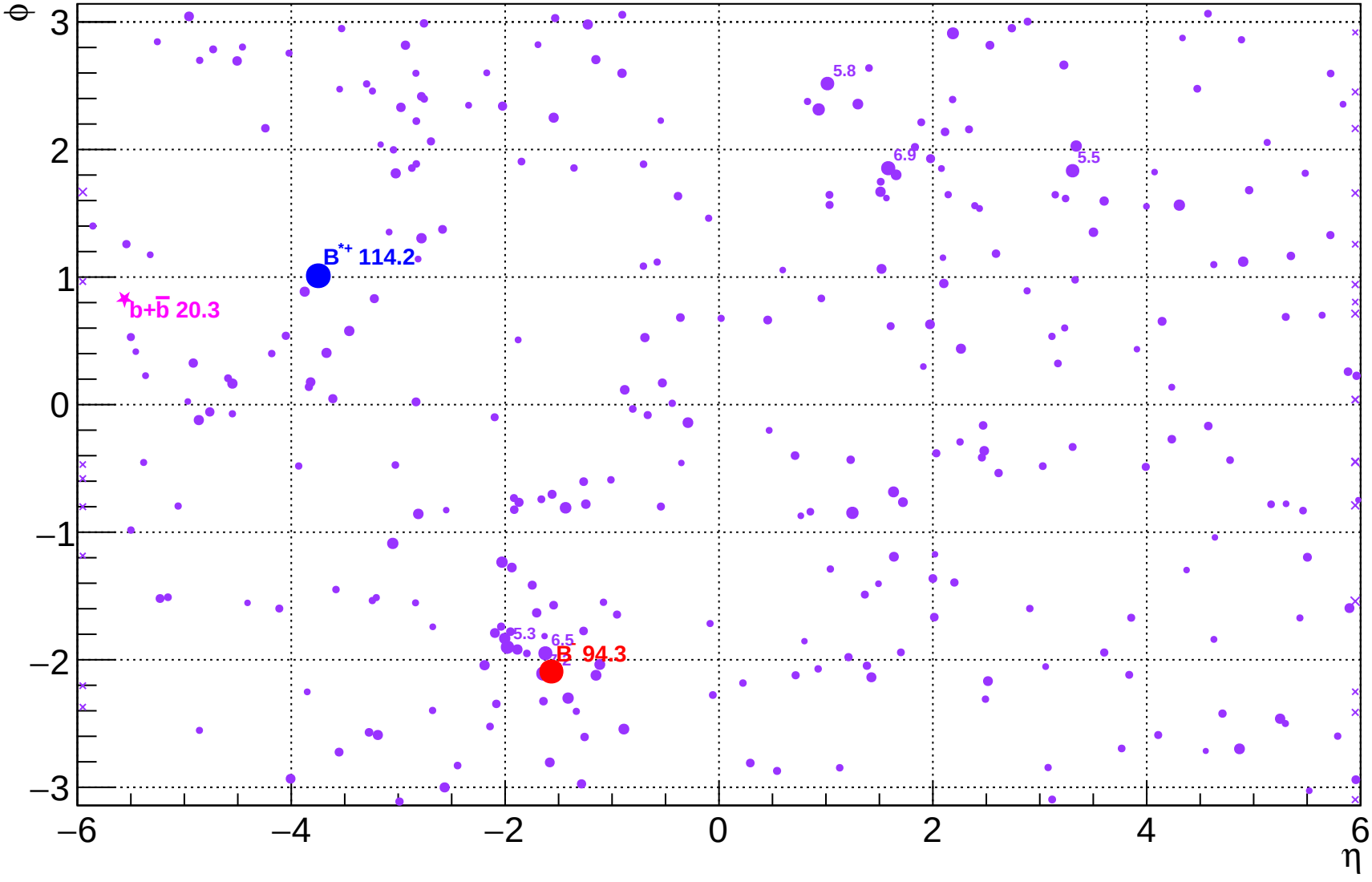
320 particles drawn (marker size ~ log p_T; label = p_T [GeV] for p_T > 5 GeV), 25 with |η| > 6 drawn on the border (x)

b hadron B^-: p_T = 94.28 GeV, η = -1.57, φ = -2.09

$\bar{b}$ hadron B^{*+}: p_T = 114.17 GeV, η = -3.75, φ = 1.01

net b+$\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 20.30 GeV (p_x = 13.73, p_y = 14.96), $\Sigma|p_T|$ = 208.45 GeV, $\Delta\phi(b,\bar{b})$ = 3.10

sample ME_PS_MPI_Had
ISR on, FSR on, MPI on
Had on, Decay off
pTHatMin 100 GeV



particle origin (number drawn)

- b system (1)
- $\bar{b}$ system (1)
- b + $\bar{b}$ (vector sum)
- primary hadron (318)

marker size:

- p_T = 1 GeV
- p_T = 10 GeV
- p_T = 100 GeV

stages:

1 hard process
2 ISR + FSR
3 MPI / remnants
4 hadronisation
5 hadron decays